Employee Attrition and Patient Satisfaction in Healthcare for a Data Science Portfolio

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***Abstract*—The purpose of this project is to establish a portfolio of work with a minimum of two completed data science projects synthesizing all prior coursework for the Master's in Data Science degree. Datasets from the healthcare sector were analyzed to investigate relevant business problems, namely employee attrition and patient satisfaction. Random Forest and Neural Network models were selected for analysis of the employee attrition dataset using the model inspection technique permutation feature importance to answer the question - Which factors have the greatest impact on employee attrition rates in healthcare? The permutation feature importance technique was applied to the neural network model as it was found to have a higher recall score. The top five features which had the largest impact on the predictive model were: overtime, age, monthly income, number of companies worked, and years at company. Tableau was then used to create an interactive dashboard using data from the HCAHPS survey to communicate patient satisfaction in hospitals across the United States for a second portfolio project. This second project answered the question - Which hospitals have the highest levels of patient satisfaction? This tool can serve as a resource for hospitals to assess how they compare to others in their region/country in patient satisfaction.**

Keywords—healthcare, employee attrition, patient satisfaction, HCAHPS, neural network, random forest

I. Introduction

Kaiser Permanente, one of the largest health plans in America, cites its mission to provide "high-quality, affordable health care" and to "improve the health of our members" [25]. Healthcare is an ever present need in society due to an aging U.S. population contributing to an increased need for healthcare with less people to provide that care [13]. The TIAA Institute reports hospitals and health systems are under major financial pressure as costs are outpacing increases in revenue [13]. Staffing shortages and burnout are leading to heightened employee turnover rates in the healthcare system [13]. Employee turnover can be very expensive costing nearly $500,000 per clinician [26] and an average of $61,110 for a bedside RN [5]. Where the corporate world concerns itself with lost productivity, healthcare must address staff attrition since the ultimate loss is insufficient patient treatment and a negative impact on patient outcomes due to inadequate staffing levels or high employee turnover [15]. For hospitals to provide high-quality care and remain financially solvent, employee attrition and patient satisfaction must be addressed since turnover can impact patient and employee satisfaction [5]. The final goal of this project is to establish a portfolio of work with a minimum of two completed data science projects to be displayed on GitHub. These portfolio projects aim to answer two questions. Project one will answer the question - Which factors have the greatest impact on employee attrition rates in healthcare? Project two will answer the question - Which hospitals have the highest levels of patient satisfaction?

II. Literature Review

Project 1

To assist human resource managers in addressing high turnover rates prior research has tested various machine learning models to predict employee attrition [14]. A systematic review of machine learning algorithms used to predict employee turnover across industries revealed Random Forest, Support Vector Machine, Logistic Regression, and Neural Networks were the most commonly studied algorithms with Random Forest proving to be the most effective [20]. The review emphasized the need for further research in improving the explainability of models and focusing on specific sectors often underrepresented in studies, like healthcare, to address the unique needs for each particular industry [20]. To address the individual needs of the healthcare industry Jessica, E. O., Lawrence, E., Hambali, M. A., & Galadima, K. R. tested the following models on employee attrition rates in healthcare: Support Vector Machine, K-Nearest Neighbor, XGBoost, Random Forest, and used SMOTETOMEK to address class imbalance [14]. They were able to improve accuracy scores compared to prior work and achieved an overall 98% accuracy for predicting employee attrition due to the application of the SMOTETOMEK technique for data resampling [14].

Research conducted in 2020 by Fallucchi, F.; Coladangelo, M.; Giuliano, R.; & William De Luca, E. tested multiple classification models against human resource data to determine which classification models would be best suited for predicting employee attrition in general and which variables were most influential in those predictions [7]. Researchers decided they were most interested in predicting the number of people who might leave the company and therefore wanted to minimize false negatives resulting in recall as the priority measure [7]. Of all the classification algorithms evaluated Gaussian Naïve Bayes was found to have the highest recall value [7]. Monthly income, age, overtime, and distance from home were the variables found to have the largest impact on predicting employee attrition [7].

In 2022 Raza, A.; Munir, K.; Almutairi, M.; Younas, F.; & Fareed, M. M. S. continued research on IBM's HR Employee Attrition Dataset [17]. After applying the SMOTE (Synthetic Minority Oversampling Technique) to address the large class imbalance, they were able to find greater accuracy scores compared to Fallucchi, F.; Coladangelo, M.; Giuliano, R.; & William De Luca, E. [17]. The four classification models Extra Trees Classifier, Support Vector Machine, Logistic Regression, and Decision Tree were analyzed to predict employee attrition and the Extra Tree Classifier was found to have the highest accuracy score of 93% [17]. Exploratory data analysis in their study shared similar results to prior work when finding the key factors of employee attrition to be monthly income, hourly rate, job level, and age

[17]. Moving forward researchers hope to see deep learning techniques applied to predicting employee attrition [17].

More recent studies have begun to apply ensemble learning techniques to machine learning models predicting employee attrition rates with much success. Chung, D.; Yun, J.; Lee, J.; & Jeon, Y. utilized stacking ensemble learning to achieve a 97.6% accuracy rate with an ensemble model using a combination of Random Forest and Artificial Neural Network [4]. Ensemble models can be classified as several different types including bagging, boosting, and stacking [4]. Random Forest is itself an ensemble model which implements a bagging approach taking multiple data samples using the bootstrap sampling method to train individual decision trees and then aggregates the results using a voting method [4]. Boosting takes multiple data samples like bagging but instead of separately training the models and aggregating later, it trains models sequentially making adjustments to each model based on the errors of the previous model [4]. Finally, stacking is an ensemble technique which uses the predicted output values from the first model as the input values for the next (stacked) model [4]. Chung, D.; Yun, J.; Lee, J.; & Jeon, Y. applied the SMOTE algorithm to their dataset to address class imbalance as in the previous study [4]. Analysis of the most influential features to predict employee attrition showed many related to satisfaction along with performance rating and monthly income [4]. A drawback of this research is the model alone cannot explain the relationships between features therefore the authors suggest methods such as partial dependency plots or logistic regression analysis be implemented moving forward to develop a better understanding of those relationships [4].

Govindarajan, R.; Kumar, N. K.; & Reddy, S. continued research into the IBM Human Resource Analytics Dataset and tested a total of five models to predict employee attrition [10]. After evaluating a Random Forest, Support Vector Machine, Decision Tree, Gradient Boosting, and a Hybrid Model the Hybrid Model was found to have the highest accuracy at 95% demonstrating the ability of multiple algorithms to combine their various strengths and achieve a higher accuracy than any single model could [10]. In their study Govindarajan, R.; Kumar, N. K.; & Reddy, S. performed feature engineering based on feature correlation to remove some of the variables from the dataset to improve prediction accuracy of the models [10]. Distance from home, employee count, employee number, environment satisfaction, hourly rate, job involvement, job satisfaction, monthly rate, relationship satisfaction, standard hours, stock option level, training times last year, and work-life balance were eliminated due to their high negative correlation with other features in the data [10]. Despite their high accuracy there are challenges with the implementation of hybrid models. Hybrid models can be complex which leads to long training times and a significant amount of computational resources [10]. These models are also not easily interpretable which could create issues in regards to fairness and accountability – a sensitive topic when specifically applied to Human Resources [10].

The last few studies reviewed solidified the choice of Random Forest and Neural Network as the chosen models to find the features most affective on employee attrition prediction models. Al-Alawi, A. I. and Ghanem, Y. A. conducted a systematic review of research applying machine learning models to predict employee attrition and found random forest to be the most commonly used algorithm and the top three features to be age, performance rating, and salary [1]. Fukui et al. also found Random Forest to be an effective model for predicting employee turnover using real data from community mental health centers [8]. T. E. Ramya and B. Sanjay compared deep learning approaches with traditional models such as Random Forest, Logistic Regression, and Gaussian Naïve Bayes finding the hybrid deep learning model to be slightly more accurate [21].

One challenge identified in the research is finding relevant datasets as human resource data is sensitive and therefore private leaving many researchers to rely on the same datasets found on kaggle.com [1]. Many also do not consider the cost-effectiveness of using the models nor do they consider the ethical implications [1]. Choosing two standalone models for this project is one way to consider the practicality of using these models in the corporate world. Karthick, K. K.; Vimalnath, V.; Prakash, N. U.; Veluchamy, R.; Kanagaraj, M. G. A.; & Sathiyamurthi, K. emphasized the need for explainability in machine learning models used in the healthcare sector which they achieved using a Hybrid Attention-Based Deep Neural Network that combines TabNet feature encoding with an Attention Residual Fully Connected Network [15]. This achieved a 96.8% accuracy and they noted many existing models struggled to correctly classify borderline cases [15]. To address this ADASYN will be used to correct class imbalances in this project.

Based on all research reviewed two models seem most appropriate for researching employee attrition in healthcare. Recent research has found a higher level of accuracy using hybrid and stacking ensemble learning models, however these can quickly become complex. For the sake of simplicity a Random Forest and Artificial Neural Network model will be used as Random Forest alone was found to have the highest accuracy for a single model according to a systematic review by Talebi, H.; Bardsiri, A. K.; & Bardsiri, V. K. [20]. Since much of the research rarely focused on newer deep learning models, Neural Network will be the second model of choice. Additionally, a combination of Random Forest and Neural Network was found by Chung, D.; Yun, J.; Lee, J.; & Jeon, Y. to have the highest level of accuracy [4]. Focusing on two individual models will allow the focus to remain on determining the features which have the greatest influence on the models.

Project 2

Data visualization alone is an incredibly important tool in analyzing the vast amounts of data industries now possess. In healthcare Srivastava, D. emphasized the ability of interactive dashboards to improve patient outcomes, give healthcare providers up-to-date information on patient metrics, and provide patients a tool to understand the implications of their lifestyle choices [18]. These same dashboards can be applied to data on hospitals themselves and used by human resources to address employee attrition. Wang, Z.; Sundin, L.; Murray-Rust, D.; & Bach, B. studied the ability of cheat sheets to communicate data visualization techniques [24]. To create these cheat sheets they were inspired by infographics, data comics, and cheat sheets in other domains to create an easy, accessible resource for different types of charts [24]. They used a real-world example to tell a story describing how to create various charts and common pitfalls to avoid while ending with a joke to promote engagement [24]. Storytelling and accessibility remain crucial when developing data visualizations and will be needed to create an interactive dashboard visualizing patient satisfaction in hospitals across the United States.

III. Exploratory Data Analysis

Project 1

The first project uses the *Employee Attrition for Healthcare* dataset from kaggle.com to analyze the various factors affecting employee attrition rates [6]. It is a synthetic dataset based on the IBM Watson dataset for attrition with 1676 records and 35 features including the target feature “attrition” [6]. There are no missing fields in the dataset and 26 of the 35 fields contain numeric values [6]. The nine fields with categorial variables include: Attrition, Business Travel, Department, Education Field, Gender, Job Role, Marital Status, Over 18, and Overtime [6]. The dataset has a large class imbalance – the target variable of attrition has 199 true values compared to 1477 false values [6].

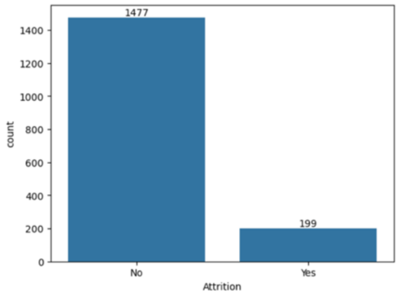


Fig 1. *Count Plot – Attrition (Predicted Variable)*

Further exploratory data analysis revealed additional imbalances in the dataset though a significantly smaller ratio than the feature – attrition. The gender field has a larger percentage of males with 998 males and 678 females. [6].

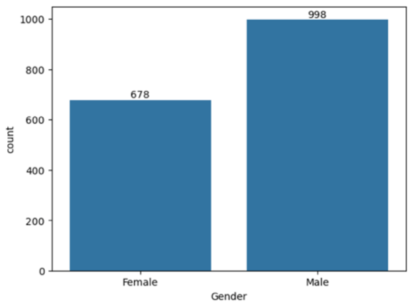


Fig 2. *Count Plot – Gender*

The spread of the data focuses mainly on nurses in one of three departments – cardiology, maternity, and neurology [6].

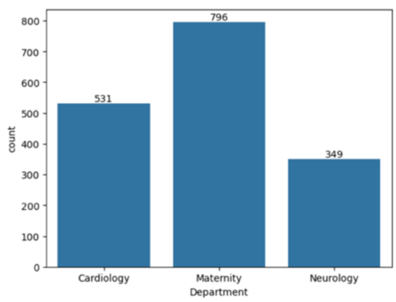


Fig 3. *Count Plot – Department*

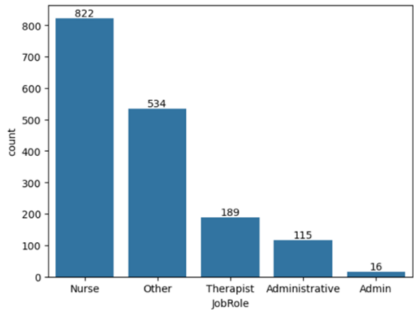


Fig 4. *Count Plot – Job Role*

A correlation of initial numeric variables was calculated and displayed as a heat map using Seaborn in Python. Most numeric variables do not show a strong correlation with each other except for variables such as total working years, job level, age, monthly income, performance rating, percent salary hike, years in current role, years at company, years with current manager, total working years, and years since last promotion.

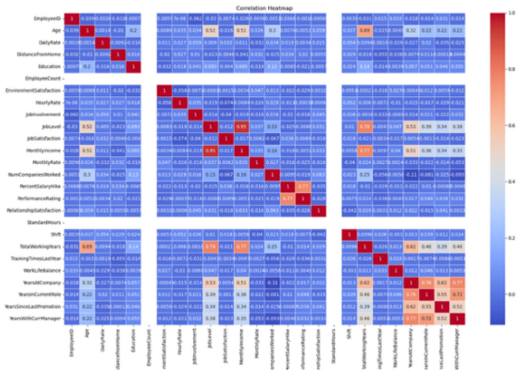


Fig 5. *Correlation Heat Map of Initial Numeric Variables*

A second csv file was included on kaggle.com intended as the test set [6]. The test set includes 206 records and only 33 fields [6]. The test set removes the attrition column since it is the predicted variable and the “Over18” column since all records were from people over 18 [6]. All fields have been converted to numeric values and normalized, though the exact methodology is not clearly stated and thus will be avoided for testing [6]. Instead, a subset of the original file will be saved for testing the chosen machine learning models.

Project 2

The second project uses data concerning patient satisfaction for hospitals across the United States which accept payments from Medicare and Medicaid [3]. This data comes from the Hospital Consumer Assessment of Healthcare Providers and Systems Survey (HCAHPS) which collects data on patient experience after a recent hospital stay [2]. The dataset to be used in Tableau to create an interactive dashboard contains 7,505,535 records with 23 feature columns for the years 2020-2025.Twenty-one fields contain categorical variables while two contain numeric values including year and zip code. The main feature which will be considered when creating the interactive dashboard is the “Patient Survey Star Rating”. The domain for this feature includes: 1, 2, 3, 4, 5, Not Applicable, and Not Available. A count plot of the dataset reveals an overwhelming majority of the data is “Not Applicable” for the Patient Survey Star Rating column. The rating 1 has 27,606 records, 2 has 112,561 records, 3 has 210,984 records, 4 has 180,915 records, 5 has 63,512 records, Not Available has 296,767 records, and Not Applicable has 6,613,190 records. Beginning in 2015 a Star Rating was added to the HCAHPS survey records as an overall rating to make it easy for consumers to quickly assess hospitals [2]. For a hospital to have a Patient Survey Star Rating they must have at least 100 completed HCAHPS surveys for a four-quarter period, thus, any hospital without a Star Rating did not meet that criteria [2].

IV. Methodology

Project 1

Data Preparation: The data was first separated into a train and test set with an 80/20 split. 35 features were identified including the target feature of attrition with 26 of the features as dtype integer and 9 as dtype object. Exploratory data analysis was performed to assess the spread of the data. After initial histograms of all integer features were performed Employee Count, Performance Rating, and Standard Hours were removed. Employee Count and Standard Hours all held the same value, and Performance Rating had nearly every row as 3.0 with a small percentage at 4.0 providing little predictive value to the models. Next, count plots of all object columns were made, and it was determined to remove the column Over 18 from the dataset as all rows contained the same value as well. Additionally, Employee ID as an identifying feature was removed since it would provide no predictive value to the models. At this point a Correlation Heatmap was created to visualize the correlation between features. Daily Rate, Monthly Rate, Job Level, Years with Current Manager, Years in Current Role, Total Working Years, and Percent Salary Hike were all removed from the dataset as they had high correlations with other features in the dataset and would provide redundant information to the models thereby decreasing performance. This was done to both the train and the test set.

Next, categorical data needed to be converted to integers for the models. For the gender column male was coded as 0 and female as 1. For the overtime column yes was converted to 1 and no to 0. For the target feature attrition yes was coded as 1 and no as 0. This was done to both the train and the test set.

At this point pipelines were built using Scikit-Learn for both the Random Forest and Neural Network models. In the pipeline the remaining categorical data will be converted to numerical values via one-hot encoding and numeric values will be normalized using the Standard Scaler in preparation for training the models. Pipelines are utilized to ensure the data is normalized based only on the training dataset to avoid data from the test set leaking into the training set. As there is a large class imbalance adaptive synthetic sampling known as ADASYN is used to manufacture synthetic samples of the minority class in this case where the target value of attrition is no [16]. What sets ADASYN apart from SMOTE is its focus on hard to classify data points whereas SMOTE creates uniform samples of the minority cases [16].

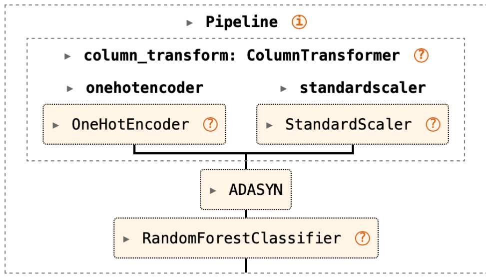


Fig 6. *Pipeline for Neural Network*

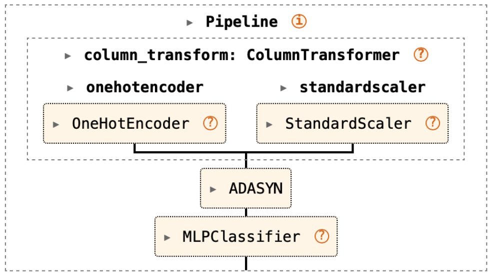


Fig 7. *Pipeline for Random Forest*

Model Selection: Random Forest and Neural Network are the two chosen models to be utilized alongside the model inspection technique permutation feature importance to determine the features most likely to impact the model’s ability to predict employee attrition in healthcare. Permutation feature importance is a model inspection technique which shuffles values and calculates how much the model’s accuracy (or other relevant score) decreases to determine how important that feature is to the model [27].

Validation: To evaluate the model recall and accuracy will be calculated. Recall is often used when predicting employee attrition as the consequences of incorrectly identifying an employee as remaining at the company is far worse than false positives [1]. Minimizing false negatives is the highest priority in healthcare HR analytics since an underestimation of attrition could lead to staffing shortages which would have a negative impact on patient care [15]. Partial dependence plots will then help visualize the effect each feature has on the model to help determine which features are most predictive of employee attrition in healthcare.

Tools: Python is the programming language of choice to code the models on a MacBook Pro computer. The libraries NumPy, Matplotlib, Pandas, Seaborn, Imbalanced-Learn, and Scikit-Learn were also used in the process.

Project 2

This project’s intended goal is to create a compelling data visualization tool to communicate patient satisfaction in hospitals across America [12]. Tableau will be the software of choice to tell a visual data story and answer the question – Which hospitals have the highest levels of patient satisfaction? This tool can serve as a resource for hospitals to assess how they compare to others in the region/country in patient satisfaction.

Data for the years 2020-2025 was sourced from the archives for the *Centers for Medicare & Medicaid Services* at Data.CMS.gov [3]. Records from the 2021 archived data snapshots through 2026 archived data snapshots were downloaded and csv files pertaining to the HCAHPS survey were selected along with the Hospital General Information csv files [3]. Data from the HCAHPS survey is published on Medicare.gov and based on the four most recent quarters of surveys. [2]. When each new quarter of surveys is updated, the oldest quarter moves to the archives [2]. Due to this there are four files for each year except for 2020, 2021 and 2025. The year 2020 only contained two csv files with records from July to December – presumably due to the Covid 19 pandemic. The year 2021 had five csv files for no explicable reason and the year 2025 only had one csv file since only the first quarter of surveys from 2025 have rolled off into the archives.

Using a Jupyter notebook the csv files for each calendar year were merged into one csv file per year. A separate column with the year was added since the only date columns were start date and end date. Next, all six HCAHPS csv files were merged into one csv file containing all data from the HCAHPS survey for the years 2020-2025. The same protocol was followed for the Hospital General Information csv file which contained a pertinent "Hospital Overall Rating" column and held data columns matching the original csv files from Kaggle. Three columns had different names depending on the year. Some years the column City was labelled City/Town, the column County Name was labelled County/Parish, and the column Phone Number was labelled Telephone Number. To ensure proper merging all applicable columns were renamed City, County Name, and Phone Number respectively.

Next, exploratory data analysis was conducted on the new dataset and cleaned to prep the datasets for visualizations in Tableau. The current HCAHPS csv files contains 7,505,535 entries with a total of 23 columns. Initially all columns in the HCAHPS file were dtype object except for the zip-code and year column which were dtype integer. The main feature of concern is the "Patient Survey Star Rating" as it is an overall score of a patient's level of satisfaction after a recent hospital stay. The "Patient Survey Star Rating" column contained the unique options of Not Available, Not Applicable, 1, 2, 3, 4 and 5. The column was then converted to dtype integer and the values "Not Available" and "Not Applicable" were all converted to null values to allow for aggregation in Tableau. This same procedure was applied to the following columns: HCAHPS Answer Percent, HCAHPS Linear Mean Value, Number of Completed Surveys, and Survey Response Rate Percent. The "Patient Survey Star Rating Footnote" column contained integer values which all correspond to a statement according to a *Hospital Data Dictionary* pdf included in all archived files at Data.CMS.gov [3]. All integer values in this column were converted to the matching statement as an option to include in the interactive dashboard on Tableau. The updated dataset was saved as a new csv file ready for data visualization in Tableau.

Four charts were chosen to create an interactive dashboard comparing patient satisfaction in hospitals across the United States. The target audience for this dashboard is hospital administrators and human resource officials to use as an investigative tool to assess how their patient satisfaction rates compare across the United States by state and local regions. A choropleth map was the first chart chosen to show the spread of data since a choropleth map allows for shading by state with the darkest states having the highest levels of patient satisfaction on average. The second chart selected is a bar chart to display the top five states by average patient star rating. The third chart chosen was another choropleth map of the average star rating by county for a selected state. Finally, the fourth chart selected was a line chart visualizing one specific hospital's average star rating from the years 2020-2025.

A layout of the dashboard was selected with the national choropleth map on the top half. The bottom half of the dashboard will be split vertically three ways. Moving left to right the bar chart, then state choropleth map, and line chart will be placed. Selecting states on the national choropleth map will select the state to populate for the state choropleth map. A year filter at the top of the dashboard will calculate the star ratings based on an aggregation of all years from 2020-2025 or one single year. A hospital filter will display the average star rating by year for a chosen hospital on the line chart. Lastly, at the bottom of the dashboard a footnote in text will appear indicating the patient experience footnote for the specific hospital chosen if available in the data.

This particular layout was chosen to emphasize the most important pieces of information for the administrators or HR personnel using it. People read from left to right and top to bottom so a Z-layout guides the user from the top left to the bottom right [23]. Dashboards should always contain the most important piece of information in the top left [23]. In this case the user must begin with the national choropleth map on the top of the graph and then second select the chosen year to populate the rest of the charts. Placing the choropleth map on the top half and making it the largest chart grabs the user's attention quickly and makes for a clear and accessible visualization tool. The line chart was placed bottom right as the second most important spot on the dashboard [23].

V. Results

Project 1

A random forest model with 100 trees was initially fit to the training data. The model was then tested using the test data and had an accuracy of 0.92 and a recall of 0.51. Next, a neural network using the MLP classifier from the sklearn library with one hidden layer consisting of 22 nodes was used for training. When tested with the test data it received an accuracy of 0.91 and a recall of 0.62. A grid search using the GridSearchCV function from the sklearn library was then used for hyperparameter tuning of the models [11]. The best estimator for the random forest model was determined to have a max depth of 15, max number of features as "log2", and 150 trees for estimation [9]. The random forest pipeline was updated and after testing was found to have an accuracy of 0.92 and a recall of 0.49.

```
Random Forest:
              precision    recall  f1-score   support

           0       0.94      0.98      0.96       297
           1       0.76      0.49      0.59        39

    accuracy                           0.92       336
   macro avg       0.85      0.73      0.78       336
weighted avg       0.92      0.92      0.92       336
```

Fig 8. *Test Report for Random Forest using Best Estimator*

Hyperparameter tuning for the neural network model found the best estimator to have one hidden layer with 22 nodes, an initial learning rate of 0.01, and max iterations of 1000. After updating the neural network model and testing, the model was found to have an accuracy of 0.92 and a recall of 0.69.

```
Neural Network:
              precision    recall  f1-score   support

           0       0.96      0.95      0.95       297
           1       0.63      0.69      0.66        39

    accuracy                           0.92       336
   macro avg       0.79      0.82      0.81       336
weighted avg       0.92      0.92      0.92       336
```

Fig 9. *Test Report for Neural Network using Best Estimator*

The next step consisted of using the permutation feature importance technique from the sklearn library to determine which features have the biggest impact on the model. The technique was run on the neural network model only as it had a higher recall compared to the random forest model. The technique was first applied using accuracy as a metric providing the results below.

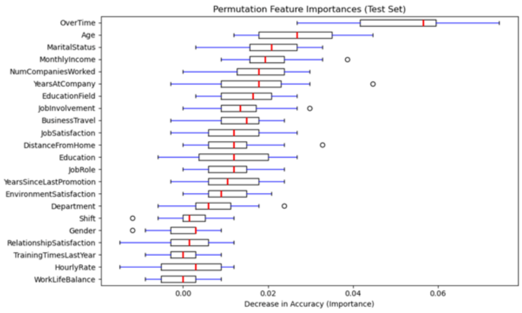


Fig 10. *Permutation Feature Importance Results - Accuracy*

Overtime, age, marital status, monthly income, and number of companies worked were the top five features which had the largest decrease in accuracy scores when the values were shuffled for that particular feature. When the test was run a second time the top five features with the largest decrease in accuracy were overtime, age, monthly income, number of companies worked, and years at company.

```
OverTime 0.057 +/- 0.011
Age      0.022 +/- 0.011
MonthlyIncome 0.021 +/- 0.010
NumCompaniesWorked 0.020 +/- 0.008
YearsAtCompany 0.018 +/- 0.011
MaritalStatus 0.018 +/- 0.008
BusinessTravel 0.017 +/- 0.005
DistanceFromHome 0.016 +/- 0.007
EducationField 0.015 +/- 0.007
YearsSinceLastPromotion 0.013 +/- 0.008
JobInvolvement 0.013 +/- 0.007
JobSatisfaction 0.012 +/- 0.007
EnvironmentSatisfaction 0.012 +/- 0.006
JobRole  0.012 +/- 0.007
Education 0.012 +/- 0.008
Department 0.007 +/- 0.007
RelationshipSatisfaction 0.004 +/- 0.004
TrainingTimesLastYear 0.003 +/- 0.006
Shift    0.000 +/- 0.005
Gender   0.000 +/- 0.006
HourlyRate -0.000 +/- 0.006
WorkLifeBalance -0.002 +/- 0.004
```

Fig 11. *Permutation Feature Importance Results - Accuracy (Mean and Standard Deviation)*

The same test was run again using recall as the scoring metric.

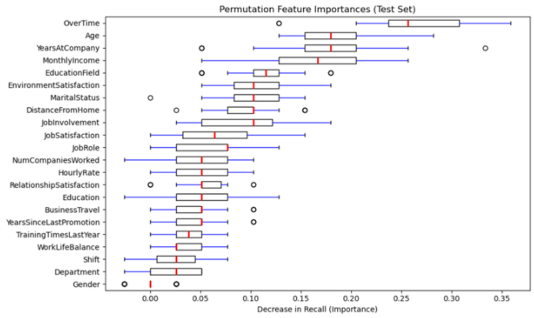


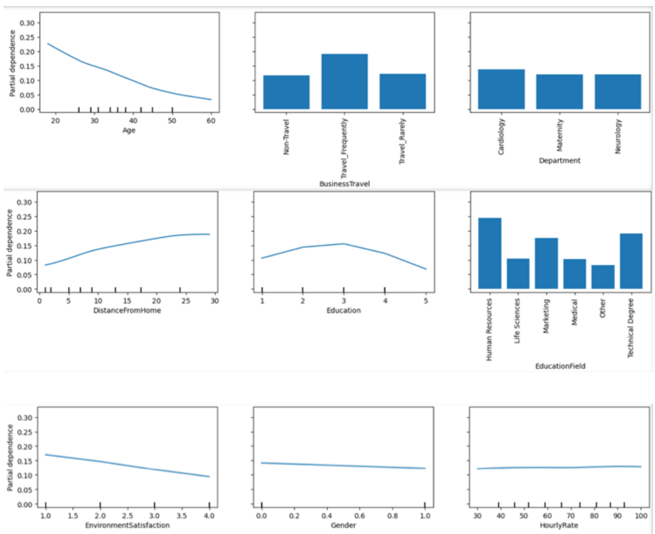
Fig 12. *Permutation Feature Importance Results – Recall*

When using recall to measure the decrease in model performance overtime, age, years at company, monthly income, and education field were the top five features which had the biggest impact on the predictive model. When run a second time the top five features were overtime, years at company, age, monthly income, and environment satisfaction.

```
OverTime 0.277 +/- 0.043
YearsAtCompany 0.188 +/- 0.055
Age      0.182 +/- 0.045
MonthlyIncome 0.160 +/- 0.037
EnvironmentSatisfaction 0.116 +/- 0.035
DistanceFromHome 0.108 +/- 0.040
EducationField 0.105 +/- 0.047
MaritalStatus 0.092 +/- 0.033
JobInvolvement 0.092 +/- 0.038
JobRole  0.070 +/- 0.036
BusinessTravel 0.061 +/- 0.022
NumCompaniesWorked 0.057 +/- 0.034
JobSatisfaction 0.056 +/- 0.040
RelationshipSatisfaction 0.051 +/- 0.021
YearsSinceLastPromotion 0.050 +/- 0.029
TrainingTimesLastYear 0.050 +/- 0.029
HourlyRate 0.050 +/- 0.022
Education 0.047 +/- 0.037
WorkLifeBalance 0.028 +/- 0.019
Department 0.022 +/- 0.029
Shift    0.020 +/- 0.027
Gender   -0.003 +/- 0.017
```

Fig 13. *Permutation Feature Importance Results - Recall (Mean and Standard Deviation)*

Finally, partial dependence plots were created using the Partial Dependence Display from the sklearn library. Partial dependence plots visually display the average effect one feature has on the predictive model's output [19].



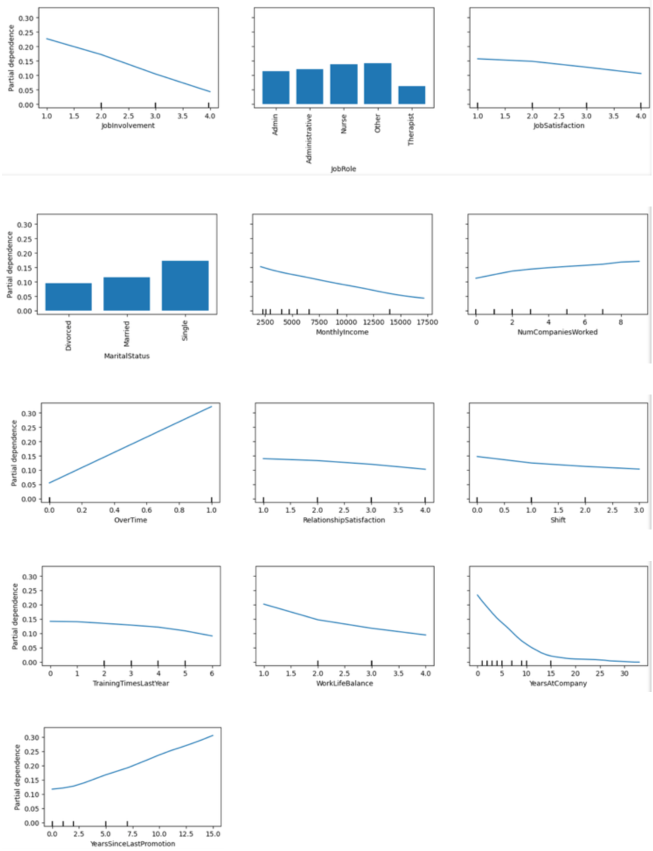


Fig 14. *Partial Dependence Plots*

Project 2

A choropleth map was created for the average Patient Survey Star Rating by state using a blue-red diverging color scheme. The states with the highest ratings are shaded in a dark blue and the states with the lowest ratings are shaded in a dark red. An interactive filter was added to change the year(s) included in the average rating. The default setting is an average of the Patient Survey Star Rating for all years from 2020-2025. When hovering over a state a window appears listing the name of the state and the numeric value of the average patient survey star rating as a number between 1 and 5.

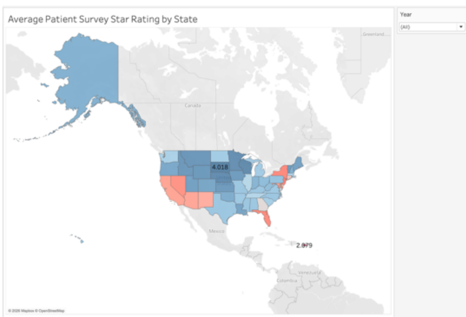


Fig 15. *Choropleth Map – Avg. Patient Survey Star Rating by State*

Next, a second choropleth map was created measuring the average Patient Survey Star Rating by counties within each state. This map is intended to provide a more granular comparison for a hospital when comparing their average Patient Survey Star Rating to their local region and the entire country. A monochromatic color scheme using blue was chosen and the counties with the highest ratings are shaded in a dark blue and the lowest ratings in light blue. Data is missing from some counties so not all counties in every state are shaded. When hovering over a county a box appears stating the name of the county and the numeric value of the average Patient Survey Star Rating.

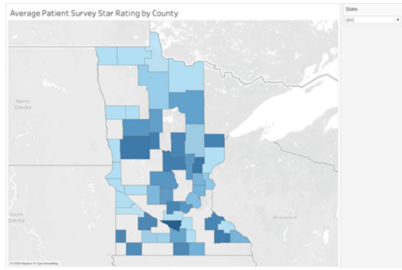


Fig 16. *Choropleth Map – Avg. Patient Survey Star Rating by County*

The current version of the national choropleth map makes it difficult to determine which states have the highest average Patient Survey Star Rating and thus a bar chart was created to help answer this question. The bar chart includes only the top 5 states by average Patient Survey Star Rating and can be filtered as an average of all 6 years from 2020-2025 or for a selected year.

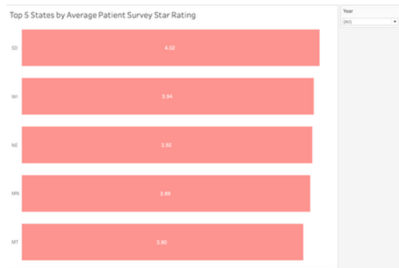


Fig 17. *Bar Chart – Top 5 States by Avg Star Rating*

The final dashboard is intended to serve as a tool for hospital administrators to see the level of patient satisfaction at their hospital and compare it to hospitals around the country. Thus, a line chart showing a hospital's average Patient Survey Star Rating for each year from 2020-2025 was created. A filter as a single-value dropdown menu to choose a particular hospital was added.

Fig 18. *Line Chart – Avg Star Rating for a Specific Hospital from 2020-2025*

The layout for the interactive dashboard was then updated and shown below.

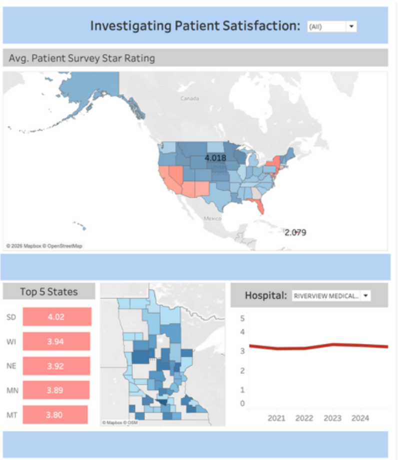


Fig 19. *Updated Dashboard*

The two most important charts in this case are the choropleth map by state on the top and the line chart in the bottom right. These charts stand out since the national map is the largest chart and the line chart is in red pulling focus. Actions were added in Tableau to make the dashboard interactive. When selecting a state in the top choropleth map the corresponding choropleth map of that state appears in the bottom middle section. When selecting a year on the top filter the bar chart, national, and state choropleth maps update based on the corresponding year. Lastly, a filter to change the hospital for the line chart was added.

VI. Discussion

Project 1

After running the permutation feature importance technique a total of four times the feature overtime was consistently found to have the biggest impact on the neural network predictive model. The features age and monthly income were found in the top five features through all tests as well. Years at company and number of companies worked were also common features typically found in the top five when running the permutation feature importance technique. Analysis of the partial dependence plots supports these findings. The plots show an increase in overtime increases employee attrition whereas an increase in age and monthly income reduces employee attrition. An increase in the number of companies worked moderately increases the chances of employee attrition whereas an increase in the years at the company significantly decreases the risk of employee attrition with the effect plateauing mostly after 15 years of work at the company.

An important consideration when using the permutation feature importance technique is the quality of the model itself. Features that appear less important for a bad model could be very important to a good model [27]. Consequently it is important to first ensure a model has good test results before running the permutation feature importance technique. While the neural network model had a relatively high accuracy at 0.92, the highest recall found was 0.69. Recall is an important metric when predicting employee attrition since underprediction could lead to understaffing which would have a direct impact on patient care along with employee and patient satisfaction. Further research should consider other deep learning techniques to produce models with higher recall rates before running the permutation feature importance technique. Additionally, partial dependence plots have limitations. Flat lines on a partial dependence plot should indicate the feature has no effect on the predictive model, however, this could be due to unknown relationships with other features in the dataset [19]. Some features might become significant when combined with one or more other features.

To address employee attrition rates in healthcare hospital administrators and human resource officials should consider the factors overtime, age, monthly income, number of companies worked, and years at the company to determine appropriate actions to reduce employee attrition. Next steps should then consider research on how different environmental and job factors work together to improve or reduce both job satisfaction and employee attrition rates. Lastly, predictive models, model inspection techniques, and partial dependence plots can never reveal the full picture of an individual employee's job satisfaction. Administrators and human resource officials should consider the factors determined here along with individual assessments of employees at their hospital/healthcare setting before making final recommendations or changes to workplace policy.

Project 2

This interactive dashboard highlighting the average Patient Survey Star Ratings for hospitals across the United States serves as an assessment tool for hospital administrators to compare their ratings from the years 2020-2025 to hospitals in their county, state, and across the United States. This project does not attempt to understand the underlying factors contributing to a high or low patient satisfaction score. This tool can be used in tandem with other analysis to provide a holistic overview of a hospital's success as measured by patient outcomes and satisfaction. Further investigation is required before deciding which factors to improve upon to raise the overall average Patient Survey Star Rating.

VII. Conclusion

The need for healthcare will only increase in the United States as the population of Americans over 65, direct contributors to the rise in chronic disease and health expenditure, is projected to increase nearly 50% by 2050 [13]. The long-term financial health of hospitals is essential to providing this population and all Americans with high-quality and affordable care. Ensuring hospitals remain financially solvent is necessary and investigating employee attrition rates in healthcare is one step towards achieving that goal and assists in keeping patient satisfaction rates high. Hospitals must continuously monitor the impact their staff and services are having on their patients. A visual data story communicating the levels of patient satisfaction in hospitals across America serves as a tool to ensure hospitals are adequately serving their communities. Visual communication tools will remain a major emphasis for addressing these relevant business problems in healthcare and providing insight into the factors most influential in predicting employee attrition. Visual charts and dashboards created using Python and Tableau help address these relevant business problems in healthcare today.

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